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Seeing the Process Through the Worker's Eyes: Event Logs from Egocentric Factory Videos

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Abstract

Manual factory work often leaves few records in information systems. We present a method and data resource for constructing event logs from egocentric video and interpreting the resulting measurements. Five stages produce transcripts, worker summaries, factory vocabularies, interval events, and merged logs. The released artifacts contain 322 per-video logs with 10,698 events from 59 workers in six factories. Extraction uses no task-specific training or manual input annotations. All released events have the required fields and permitted labels, but these properties do not establish accuracy against video. An existing author audit finds full transcript support for 36 of 42 sampled events, partial support for six, and no contradictions. We formalize the representation, document temporal inconsistencies, and reproduce the existing descriptive calculations from frozen outputs. These calculations assign 10.8% of retained summed event duration in one factory to defect patching, a behavior already described in its report. They also show why a zero transport category can coexist with 101 recorded transport events: categories are assigned from process context. Worked examples, metric definitions, and an analysis of missing time explain what these logs can and cannot establish. The resource supports inspection and hypothesis generation; video accuracy, causal diagnoses, and intervention effects remain unmeasured.

Keywords: process mining; inferred event logs; egocentric video; measurement; data quality; manufacturing; foundation models

1. Introduction

Process mining uses timestamped event records to study operational processes [1]. Many manual factory tasks leave no such records. An information system may record an order or machine checkpoint while missing how a worker loads parts, moves material, checks a component, or switches tools. Video can make some of this work observable, but it does not directly provide the case identifiers, labels, and event boundaries that process mining needs [2].

Egocentric video records work from the camera wearer's perspective. The public Egocentric-10K resource provides factory recordings of this kind [3]. Figure 1 illustrates the viewpoint. Turning recordings into event logs requires decisions about what constitutes an event, which names to use, and how to interpret time. These are measurement decisions as well as extraction tasks. A well-formed log can still misrepresent the work if a behavior is omitted, a boundary is estimated poorly, or an analytical category hides a recorded action.

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Figure 1. Example frames from Build AI's Egocentric-10K resource [3]. They illustrate the camera viewpoint, rather than the accuracy of the derived event labels.

We study a five-stage pipeline that produces timestamped transcripts, worker summaries, factory reports and vocabularies, per-video events, and merged logs. Its application covers six factories, 59 worker identifiers, and 322 recordings. The resulting 10,698 events are inferred operational episodes. They do not provide a complete census of production cycles or manufactured items.

This journal revision treats the pipeline and its released outputs as a methods and data resource. Its contributions are: a formal definition of interval events and their provenance; an implementation account that separates prompt requests from software checks; an assessment with explicit denominators and traceable examples; and a measurement analysis of abstraction, missing time, and dependent evidence. These additions use the existing outputs and author judgments. They add no model runs, annotations, datasets, ablations, or discovery settings. Appendix A distinguishes new methodological substance from fuller reporting of the existing study.

Four research questions organize the paper. **RQ1:** How are the inferred logs constructed, and which structural properties do the released records satisfy? **RQ2:** How well do event timing and meaning agree with the generated transcripts? **RQ3:** Which descriptive analyses do the logs support, and what limits their operational interpretation? **RQ4:** What general measurement lessons follow from the representation, temporal rules, and shared evidence? Table 1 links each question to its evidence.

Table 1. Research questions, evidence, and limits. The full revision claim register is released in `paper/revision/claim_evidence.csv`.

Question and section	Supporting artifacts	Finding	Main limit
RQ1, Sections 4 and 7.1	Scripts, prompts, reports, all per-video CSVs	Fields, ordering, and vocabulary membership can be checked	Valid structure does not imply correct observation
RQ2, Section 7.2	Transcripts, logs, 84 archived audit judgments	Partial temporal and semantic consistency	No independent video reference; timestamp anomalies
RQ3, Section 7.4	Frozen events, process-category map, reports	Composition, sequence, and worker-profile descriptions	Unequal windows, task mixes, and category meanings
RQ4, Section 8	Worked examples, merge code, metric definitions	Aggregation can hide behavior; missing time has two-sided effects	Conceptual deductions do not estimate unknown errors

2. Background and Related Work

2.1. Constructing and assessing event data

Diba et al. [4] distinguish extraction, correlation, and abstraction when constructing event data. These tasks also arise here: video must become a description, episodes need labels, and a case rule must group them. The abstraction review by van Zelst et al. [5] explains how grouping low-level events changes the level at which a process can be studied. In our setting, a repeated sequence can already be compressed during narration, before a structured event exists. Consequently, a row count cannot be assumed to count cycles.

Suriadi et al. [6] describe recurring event-log imperfections and their implications for analysis. Their work motivates checks for missing fields, overlapping intervals, and inconsistent time representations. These checks expose data problems but do not establish that a visually inferred activity is correct. We therefore separate structural checks, vocabulary checks, transcript consistency, and accuracy against footage.

2.2. Video and sensor methods

First-person workflow extraction without task-specific labeled training data predates this study. Pham and Xiao [7] extract mechanical-assembly workflows from first-person video without pre-labeled training data or pre-trained classifiers. Their preliminary evaluation already addresses the combination of egocentric observation and unsupervised extraction. Knoch et al. [8] use hand trajectories and clustering at a camera-equipped assembly workstation, evaluated with ten participants. Kratsch et al. [2] develop a reference architecture for video-based process mining. Lepsien et al. [9] combine object tracking, action detection, and abstraction to produce event data. The sensor-data review by Brzychczy et al. [10] addresses the related problem of turning lower-level signals into process events. These streams inform our separation of observation, labeling, and case construction.

SnapLog [11] combines image embeddings, temporal segmentation, and a classifier adapted with labeled examples. It evaluates both TUM Kitchen recordings and 24 egocentric EPIC-KITCHENS videos from ten environments. It can retain distributions over candidate activity labels. Thus, prior work is not uniformly restricted to fixed cameras, one workstation, or one supervision regime. Table 2 makes the closest distinctions explicit. The present contribution is the combination of a multi-factory application, induced factory vocabularies, inspectable text, a layered artifact release, and an account of operational measurement. No comparative accuracy advantage is claimed.

Table 2. Comparison of the closest video approaches. "Not established" means that the inspected source does not establish that property; it is not a claim of absence.

Dimension	Pham and Xiao [7]	Video-to-Model [8]	SnapLog [11]	This resource
Viewpoint	First-person	Camera at assembly workstation	Includes egocentric video	Egocentric factory video
Supervision	No pre-labeled data or pre-trained classifiers	Unsupervised clustering of hand trajectories	Pre-trained encoders; labeled adaptation examples	General-purpose models; no task training or input annotations
Label induction	Steps and operations with semantics	Clusters assigned to work steps	Adapted activity classes	Text-induced activity and process lists per factory
Intermediate representation	Extracted workflow semantics	Hand trajectories	Embeddings and segments	Transcripts, worker summaries, factory reports
Uncertainty	Calibrated event uncertainty not established	Calibrated event uncertainty not established	Candidate-label probabilities	Textual caveats; no event confidence field
Domain	Mechanical assembly	Manual assembly	Kitchen activities	Six manufacturing settings
Evaluation scope	Preliminary feasibility	Ten participants in a laboratory study	Two datasets; multiple environments	322 logs; transcript checks and descriptive analyses

2.3. Language models, uncertain data, and manufacturing interpretation

LLMs have been studied as assistants for process mining [12] and for generating database queries used in event-log extraction [13]. Kecht et al. [14] address log construction from customer-service conversations using natural language inference. Here, the text is itself model-generated evidence. A semantic check against that text tests one link in the extraction chain, not the entire path from physical work to an analytical conclusion. This dependency is central to our assessment.

Work on uncertain event data makes uncertain labels, times, or event occurrence explicit [15]. Our deterministic CSVs do not implement such an uncertainty model. Readable intermediate text helps a reader investigate an event, but it does not provide calibrated probabilities or formal bounds. Similarly, object-centric process mining [16] clarifies why worker and recording identifiers cannot substitute for product identifiers when tracing material across stations. These distinctions limit claims about throughput, yield, handoffs, and complete manufacturing instances.

Industrial process-mining research also spans several analytical goals, as reviewed by Corallo et al. [17]. This motivates separating a description of recorded work from a decision about changing production. Our time-composition and sequence measures provide evidence for the former; the latter requires product, quality, and operational context absent from the inferred logs.

3. Problem Definition

Let F be a set of factories, W_f the worker identifiers associated with factory f, and V_w the recordings associated with worker w. Each video v has an underlying duration D_v, which need not be available in the derived archive. A transcript T_v is an ordered collection of entries (s_i, e_i, x_i), where x_i describes the interval from s_i to e_i. These endpoints and descriptions are model outputs. They are not independent annotations of the source video.

For each factory, the pipeline induces an activity vocabulary A_f and a process vocabulary P_f. An interval event is

$$e = (v, w, f, t_s, t_e, a, p), \qquad a \in A_f, \quad p \in P_f. \tag{1}$$

The tuple makes provenance explicit: the CSV's path identifies v, while w and f are stored in each row. The estimated endpoints t_s, t_e are relative to the video. A case function $\kappa : E \to K$ groups events into traces; it is an analytical choice. The construction problem is to derive E, A_f, and P_f from the videos while retaining enough source information to inspect and interpret the result.

Table 3. Notation and intended meaning.

Symbol	Meaning
f, w, v	Factory, worker, and video identifiers; identifiers do not certify roles or task equivalence
D_v, T_v	Source duration and generated transcript; the former is unavailable in the released subset
A_f, P_f	Two separate permitted label lists for factory f
t_s, t_e, d_e	Estimated video-relative start, end, and duration $d_e = t_e - t_s$, in seconds
$\kappa(e)$	Selected case identifier, such as the source video
$g_f(p)$	Analyst's mapping from process label to operational category
C_v, S_v	Interval-union event duration and endpoint span used in coverage diagnostics

An event represents an operational episode. One episode may summarize many repeated actions, and several events may divide a single narrated interval. It need not be one item, one production cycle, or one complete process instance. Desirable properties

include faithful descriptions, meaningful boundaries, appropriate labels, and sufficient coverage. The implementation enforces only a subset: parseable offsets, end not before start, required values, and membership in the separate vocabularies. Section 4.2 distinguishes these checks from intended properties.

3.1. Activity, process context, and analytical category

An *activity* answers what action is performed. A *process context* identifies the broader operation to which the extractor assigns it. An *operational category* answers how an analyst groups the action or context for a particular question. These concepts are separate. The implementation checks $a \in A_f$ and $p \in P_f$ independently. It does not enforce a compatibility relation $H_f \subseteq A_f \times P_f$ or a process-to-activity hierarchy.

The current time analysis uses $g_f(p)$, not an activity-based mapping. Thus, transport within surface finishing can receive the category assigned to finishing. Figure 2 gives an exact released example. An explicit compatibility relation or a category rule using (a, p) could improve future analyses, but neither was enforced in this extraction. Reclassifying all activities would define a different analysis and is not part of the present revision.

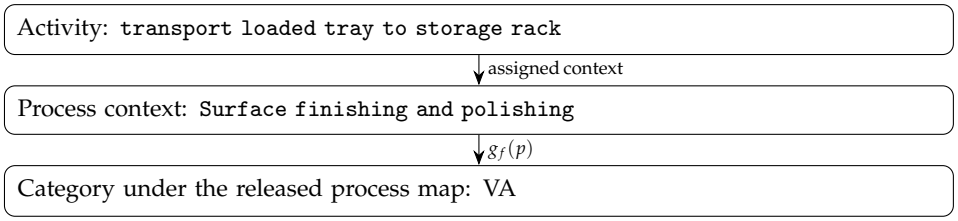


Figure 2. A transport action can contribute to VA time under a process-based category rule. Source: factory 003, worker 001, recording 00001, CSV data row 2, from 00:03:52 to 00:04:00 (8 seconds). The action is recorded; the category aggregation hides it.

4. Pipeline and Design Rationale

The released scripts implement five stages. S1 uses video and a transcription prompt. S2 combines a worker's transcripts. S3 combines a factory's worker summaries. S4 uses the original transcript again, together with the factory vocabularies and the compact worker entry from the S3 report. S5 merges the resulting CSVs. Figure 3 shows these dependencies. Reports and logs share evidence, so their agreement is not independent validation.

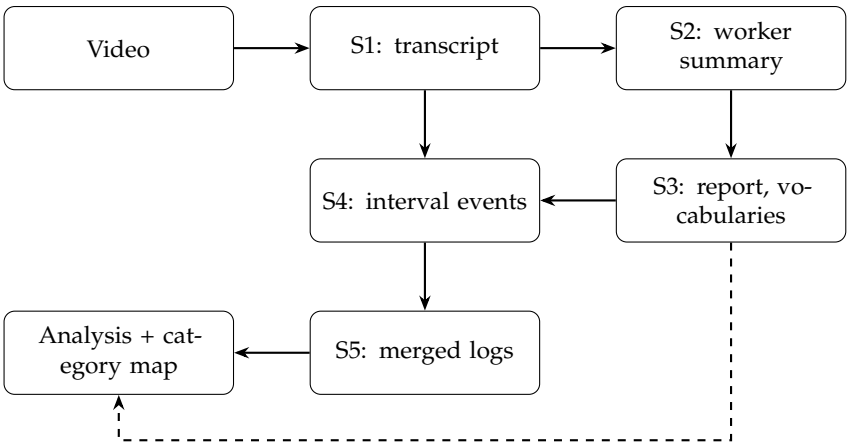


Figure 3. Evidence dependencies in the implemented pipeline. S1 to S3 use Gemini requests; S4 uses a text-only OpenAI request. The analysis map is an author-defined interpretation. An error in the transcript can affect both the narrative report and the extracted log.

4.1. Execution units and failure behavior

Table 4. Stage specification from scripts 01 to 05. A nonempty text output is not a validation of its factual content.

Stage; unit	Inputs and context	Output	Failure and reuse behavior
S1; video	Uploaded video; transcription prompt	One chronological text file	Keeps existing files; retries transient failures; rejects an empty response; attempts deletion of uploads
S2; worker	All saved transcripts in one request; filenames omitted from the bundle	Worker report	Missing or empty input raises an error; empty response rejected; keeps existing output unless overwrite is set
S3; factory	All worker summaries in one request, with worker identifiers	Factory report with two label lists and compact worker entries	Same text and reuse checks as S2; malformed section structure may fail later in S4
S4; transcript	Transcript; two label lists; compact S3 worker entry, or a missing-context marker	Per-video CSV	Missing lists or invalid events raise errors; an empty event list is valid; existing CSV kept unless overwrite is set
S5; worker, then factory	CSVs in sorted filename order	Worker and factory CSVs	Checks timestamp columns and consistent headers; empty files add no time; keeps existing output if its schema matches

```
for each selected factory f:
    for each available video v:
        T[v] = reuse existing transcript or S1(video[v], prompt1)
    for each worker w:
        B = concatenate all T[v] for w, without filenames
        R[w] = reuse existing summary or S2(B, prompt2)
    F[f] = reuse existing report or S3(all R[w] for f, prompt3)
    A, P, worker_context = parse the required sections of F[f]
    for each transcript T[v]:
        context = T[v] + A + P + worker_context[w]
        E[v] = reuse CSV or normalize(S4(context, prompt4))
    for each worker w:
        offset = 0
        for each CSV v in sorted filename order:
            append E[v] shifted by offset; keep local offsets
            if E[v] is nonempty: offset += max end(E[v])
    concatenate worker logs into the factory log
```

Listing 1: Algorithm 1: implemented dependency order. API retries surround requests; this pseudocode does not add windowing or new validation.

S1 asks for a new timestamped entry when an action, object, location, or interaction changes. It asks for repeated stable work to be summarized once and for uncertainty to be stated briefly. These are instructions to the model. S1 does not check whether each visible change was narrated or whether the timestamps match the video.

S2 and S3 compress large text bundles into summaries. This makes factory-wide vocabulary induction possible, but uncommon actions can be lost. The implementation does not automatically split long requests into windows, reconcile separate summaries, or check coverage against all transcript spans. S4 partly bypasses this compression by reading each original transcript, although it must still use the induced vocabularies.

S4 requests a strict JSON schema with estimated start and end times, activity, and process. Its local parser can also accept a list, strip code fences, recognize fallback time

keys, and parse two-component offsets or numeric seconds. It normalizes label case and surrounding quotes, checks membership, rejects end-before-start intervals, and sorts rows by start, end, activity, and process. It allows zero-duration events and overlaps. It does not check video bounds, full temporal coverage, the truth of an activity, or compatibility between the activity and process. The archive contains CSV outputs, not the original S4 JSON responses.

S1 to S3 default to `gemini-3.1-pro-preview`; S4 defaults to `gpt-5.4`, requests high reasoning effort, and permits up to 100 concurrent extraction tasks. These are recoverable code settings. The existing manuscript identifies the corresponding model families, but per-request execution records are unavailable. No temperature, token limit, seed, cost, or runtime is inferred from missing records. Appendix B specifies the distinction between code defaults and historical execution evidence.

4.2. Requests, guarantees, observations, and expectations

Table 5. Four levels of evidence for pipeline properties.

Level	Example	What can be concluded
Prompt instruction	Mark uncertainty; include handling and waiting when visible; split meaningful changes	The behavior is requested. Compliance and completeness require separate assessment.
Software constraint	Parse event fields; canonicalize permitted labels; reject negative intervals; sort rows	Accepted outputs satisfy these checks. Semantic correctness and accurate time remain open.
Observed artifact property	All 10,698 rows have required values and permitted labels; 35 adjacent overlaps are detected	These statements hold for this release. They are not promises about future runs.
Untested design expectation	Summaries help vocabulary consistency; closed labels help interpretation	These are rationales. No ablation or comparative benefit estimate is available.

"Automated extraction" means that no task-specific training or manual input annotation is required by the extraction stages. It does not mean that the complete study is free of human judgment. One author conducted the archived transcript audit, and the operational-category map reflects author choices. "Constrained" refers to the separate permitted label lists. "Validated" is used only with a named check, never as a general claim of visual accuracy.

4.3. Design choices and tradeoffs

Table 6. Design rationale and conceptual alternatives. No alternative in the last column was experimentally compared here.

Choice	Intended benefit	Limitation	Possible alternative
Video-to-text intermediate	Readable evidence; reuse for different analyses	Visual errors become shared textual evidence	Direct video-to-events, with stronger visual provenance
Worker summaries	Compact factory-level context	Rare behavior and recording identity may be lost	Window summaries with explicit source references
Factory vocabularies	Concrete local labels across workers	Different factories use different abstractions	Global concepts with local extensions
Closed labels	Prevent out-of-list output	Closest allowed label may still be wrong	Open labels followed by review and reconciliation
Change-point narration	Compact description of stable behavior	Boundary detail and repeated cycles can disappear	Finer segments with explicit repetition counts
Interval events	Preserve approximate episode duration	Row counts do not equal production volume	Object-linked cycle records, when supported

Repeated cycles deserve particular care. If ten cycles become one narrated interval and one event, the event count decreases and the directly-follows relation loses nine possible repetitions. A longer event can represent more repetitions, slower work, or more aggressive compression. The logs alone do not distinguish these explanations. The text interface makes this decision inspectable, but does not recover details omitted during transcription.

5. Dataset and Released Artifacts

5.1. Selection and provenance

The archived subset contains outputs for factories 001 to 006, with 59 worker identifiers and 322 paired transcript and event files. Factory and worker identifiers are inherited from the source organization. Domain descriptions in this paper summarize generated factory reports; they are not independently verified factory metadata. The archive does not record the selection rationale, the original source revision, or a complete inventory of the recordings available at extraction time. We therefore describe an available processed subset, not a random or representative sample of factories.

The current provider card reports a median clip length of 180 seconds [3], whereas many archived transcript endpoints lie near 20 minutes. These describe different quantities and possibly different snapshots or subsets. The card documents a per-video `duration_sec` field inside source archives, but those metadata files are not included here. The pipeline scans video files and does not preserve that field. We cannot verify the earlier claim that these selected source recordings were typically 20 minutes long or occasionally many hours long. The public source revision inspected during this revision is recorded in `paper/revision/source_metadata.json`; it must not be substituted for the unknown historical revision.

Table 7. Artifact inventory. Counts come from the frozen repository outputs. Worker and factory merged logs repeat the same source events.

Layer	Files	Content and provenance
Transcripts	322	One text per video; recording identity in path
Worker summaries	59	One text per worker; combines transcripts
Factory reports	6	Narrative, worker entries, activity and process lists
Per-video logs	322	10,698 interval-event rows; six stored columns
Merged worker logs	59	Same 10,698 events with source file, worker case, and local offsets
Merged factory logs	6	Same 10,698 events concatenated across workers
Audit record	1	42 event judgments and 42 transcript-entry judgments
Prompts; stage scripts	4; 5	Extraction instructions and implementation

The six per-video columns are `start_timestamp`, `end_timestamp`, `activity`, `process`, `factory`, and `worker`. S5 adds `source_file`, `case_id`, video-relative start and end offsets, and duration. Table 7 and the hashed artifact manifest distinguish file counts from unique event counts. The dataset card in `paper/revision/DATASET_CARD.txt` follows the documentation concerns of Datasheets for Datasets [18]: composition, collection, uses, and limitations.

There are 13,304 entries recognized by the original three-component timestamp parser. Recognizing the additional two-component and mixed-format entries gives 13,462 entries, without rescaling any timestamp. The original endpoint-span calculation totals 786,412 seconds (218.4 hours). This is a sum of parsed spans, not verified footage duration. Across all logs, interval unions cover 414,886 seconds and summed event durations total 421,283 seconds. Their difference, 6,397 seconds, reflects overlaps. The retained time-analysis set contains 10,647 events in 320 logs, with 349,659 seconds of summed event duration (97.1 hours). These quantities must not be interchanged.

5.2. Temporal data quality and inclusion rules

Table 8 identifies all seven transcripts whose literal parsed endpoint exceeds one hour. An endpoint of `20:06:00` paired with an event endpoint of `00:20:06` is consistent with a format error, but does not prove which timestamp is correct. Some files switch from `HH:MM:SS` to `MM:SS`. No timestamps are repaired in this revision.

Table 8. Transcript and event endpoints for the seven literal transcript spans over one hour. The final column applies the original time-analysis rule to event endpoints. IDs abbreviate factory/worker/video. Source durations are unavailable for all rows.

ID	Transcript endpoint	Event endpoint	Time-analysis decision
001/001/00004	20:00:00	20:00:00	Exclude
001/010/00001	01:20:00	01:20:00	Exclude
002/002/00008	20:00:00	00:20:00	Retain
002/010/00001	20:00:00	00:20:00	Retain
005/002/00000	13:52:00	00:13:52	Retain
006/001/00001	20:06:00	00:20:06	Retain
006/004/00000	19:59:00	00:19:59	Retain

The original value-analysis code excludes a log only when its maximum event endpoint exceeds 3,600 seconds. It excludes two logs, both in factory 001. The earlier prose described seven exclusions based on transcript spans; that description did not match the calculation. We preserve the implemented rule and the reproduced time tables, while stating that it leaves five of the timestamp-anomalous transcripts' logs in the analysis. Retention is a calculation rule, not certification of their timestamps.

Four other transcripts contain two-component endpoints. Their identifiers are 001/004/00001, 002/005/00001, 002/009/00003, and 003/003/00003. Three change format within the file; 002/005/00001 uses `MM:SS` throughout. The original parser misses 158 entries across these files. The inclusion manifest records every video's formats, format-transition positions, endpoints, row count, union duration, summed duration, and decision. A separate interval manifest preserves the original endpoint strings. The low coverage ratios in Section 7.2 therefore diagnose inconsistent representations; they do not establish long-video truncation.

5.3. Synthetic time and case selection

S4 writes offsets as dates relative to 2000-01-01. This date exists for software compatibility. The archive provides no verified calendar times or cross-worker synchronization. S5 processes clips in filename order, which is not proof of their real chronological order. For a worker's ordered clips $v_1, \ldots, v_m$, define $M_j = \max_{e \in E_{v_j}} t_e$, with $M_j = 0$ for an empty log. S5 uses

$$O_1 = 0, \qquad O_{j+1} = O_j + M_j, \qquad (t'_s, t'_e) = (O_j + t_s, O_j + t_e). \tag{2}$$

The next clip starts at the largest extracted endpoint, not at the full source duration. Unrepresented trailing footage and empty logs add no elapsed time. Within-clip gaps before that endpoint remain on the synthetic axis, but gaps between recording sessions are unknown. Each worker has a separate synthetic origin; equal synthetic dates do not establish simultaneous work.

6. Assessment Methodology

The assessment uses frozen texts, reports, CSVs, and archived author judgments. The descriptive workflow runs offline with Python's standard library. The original activity-discovery configuration is repeated separately with PM4Py [19], without changing the

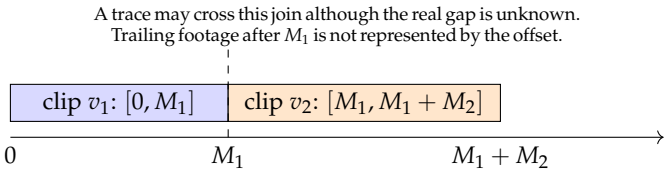


Figure 4. Schematic of the implemented merge, not measured source time. Separate video cases preserve the clip boundary. A worker case creates a directly-follows relation across it.

Table 9. Case constructions supported by the attributes. Process episodes require an additional grouping step and are not stored as case IDs.

Case rule	Supported description	Interpretation risk
Video as case, $\kappa(e) = v$	Local activity order within an observation window; used for discovery	Window may start or end midway through a production process
Worker as case, $\kappa(e) = (f, w)$	Sequence of retained clips; stored S5 case ID	Filename joins can create artificial transitions; calendar duration is unknown
Contiguous process episode	Consecutive events with the same process within a chosen clip, with a stated gap rule	Inferred context boundaries do not identify items or completed instances

miner or its noise threshold. This is recovery of an existing calculation, not a new discovery experiment. No source artifacts are overwritten.

6.1. Metrics, denominators, and aggregation

For each video, let E_v be its events and B_v the set of transcript endpoints recognized by the original parser. Let $T_v^{\max}$ be the maximum such endpoint and M_v the maximum event endpoint. The original coverage diagnostic uses

$$C_v = \left| \bigcup_{e \in E_v} [t_s, t_e) \right|, \qquad S_v = \max(T_v^{\max}, M_v), \qquad \text{coverage}_f = \frac{\sum_{v \in f} C_v}{\sum_{v \in f} S_v}. \tag{3}$$

The denominator begins at zero. This is an endpoint-span ratio, weighted by span, not a mean of clip ratios and not footage coverage. When no transcript endpoint is recognized, the event endpoint supplies the denominator. The corrected parser's results are released as a separate diagnostic; the original table is retained and labeled.

A boundary matches if its distance from *any* recognized transcript endpoint is at most five seconds. Both ends of every event enter the denominator, including events in files with no recognized transcript entries. Multiple event boundaries can match the same transcript boundary. A match need not identify the correct action or justify an internal split.

For category analysis, the process lists have 113 factory-specific entries and 108 distinct strings across factories. Of the 113 entries, 112 are used in the logs. The unused factory 005 label *motor assembly finishing* has no category assignment in the original map. Every process string actually assigned to an event is mapped. Appendix D lists the map by factory and marks the unused entry explicitly.

The categories are value-adding production work (VA), material handling and staging (MH), transport (TR), quality inspection (QI), cleaning and maintenance (CM), rework (RW), and documentation (DOC). VA includes packaging and other completion work under the authors' scheme; it is not a direct measurement of customer value. Defect patching is assigned RW because its label and descriptions concern filling or repairing surface defects. This does not establish that it is avoidable or that the upstream process failed a specified quality standard. Supporting work can be necessary even when assigned outside VA.

Table 10. Metric definitions. E_f^* is the retained set under $M_v \leq 3600$. E_w^* is its worker subset, $H_w = \sum_{e \in E_w^*} d_e / 3600$, and R_w^{VA} is the worker's list of VA run durations.

Quantity	Definition	Unit, aggregation, exclusions
Label conformance	$\#\{e : a \in A_f,\ p \in P_f\} / \#E_f$	Fraction of all event rows
Vocabulary use	Distinct assigned labels divided by permitted labels	Separate activity and process lists, per factory
Coverage diagnostic	$\sum C_v / \sum S_v$	Fraction of parsed endpoint span; all logs
Boundary agreement	$\sum_{e,b \in \{t_s,t_e\}} \mathbf{1}[\min_{t \in B_v} \lvert b - t \rvert \leq 5] / (2\#E)$	Fraction of boundaries; all logs; original parser
Category share	$q_{fc} = \sum_{e \in E_f^*} d_e \mathbf{1}[g_f(p_e) = c] / \sum_{e \in E_f^*} d_e$	Fraction of summed event duration; overlap counted in each row
Worker VA share	Same ratio restricted to E_w^* and category VA	Eligible workers require $H_w \geq 0.5$; factory median weights workers equally
VA transition rate	I_w / H_w, where I_w counts VA-to-nonVA transitions in concatenated run lists	Transitions per summed event-hour; factory median
Mean VA run	$\overline{R}_w = \sum R_w^{VA} / \#R_w^{VA}$	Minutes; table reports median across eligible workers of these means
MH/TR count and rate	Number of retained MH or TR rows; count divided by summed event-hours	Rows, not merged episodes; duration-weighted factory denominator
Dominant process	$\arg\max_p \sum_{e \in E_w^* : p_e = p} d_e$	Groups workers; comparisons then use all their retained events
Self-loop share	Equal adjacent activity labels divided by all adjacent pairs	Within video cases only; all logs

Runs collapse consecutive same-category rows *within each video*. Their duration is the sum of member durations, which can bridge unrepresented gaps and count overlaps. Run lists are then concatenated across a worker's clips before VA-to-nonVA transitions are counted. This can count transitions across a clip join; the release contains 15 such transitions among eligible workers. These rates are therefore representation-dependent transition measures, not verified interruptions of continuous work. The original script rounds each worker's VA share, transition rate, and mean run duration to one decimal before taking factory medians. We retain this rounding rule. No eligible worker has an empty VA-run list in this release; the original implementation would report zero for that case.

All per-video logs are nonempty. Negative and zero durations are checked separately; no such rows occur. Overlaps remain in time composition and are merged only for coverage. Ambiguous transcript timestamps remain literal. This preserves the original calculations while making their limits visible.

6.2. Existing semantic audit protocol

The archive records 42 sampled events and 42 sampled transcript entries, seven of each per factory. The earlier manuscript attributes the judgments to one author. The judgment file states `random.seed(42)` and describes factory stratification. However, the referenced sampling block is absent from the archived assessment script. The sampling order, full sampling frame, and handling of timestamp formats cannot be reconstructed. We report the existing sample as documented and do not claim that it can be redrawn exactly.

For the event sample, the reference is the generated text overlapping the event interval. Code C means fully supported: the label describes the transcribed action in that interval. P means partially supported: the label is plausible but the overlapping text alone is insufficient. X means contradicted. Strict support is $36/42 = 0.857$. The six P cases are retained in the denominator, not treated as fully correct.

For the transcript-entry sample, COV denotes overlap with at least one event, EXC an uncovered entry judged non-operational, and MISS an uncovered entry judged operational.

The archive reports $34/(34+4) = 0.895$ after excluding four EXC entries. We call this the *reported entry-overlap rate*, not action recall. One COV entry describes camera tilt covered by a longer event. Thus, the denominator does not establish 38 independently verified operational entries, and overlap does not show that all actions or labels were extracted correctly.

The audit measures consistency with transcripts. It cannot identify visual errors, missed actions, or mistaken object interpretations already present in the transcripts. There is no independent evaluator, inter-rater agreement estimate, or source-video audit. The sample proportions do not bound error in downstream time shares, category assignments, observation selection, or causal interpretation. Appendix E reproduces all judgments and their notes, with the unverified truncation explanation clearly marked.

7. Results and Factory Case Studies

7.1. RQ1: Construction and structural properties

Table 11 reports the released corpus. All 10,698 events have the required values, nonnegative ordered endpoints, and labels in the relevant factory lists. Start times are nondecreasing within each log. There are no empty logs or zero-duration events. However, 35 adjacent event pairs overlap, 20 in factory 001 and 15 in factory 002. Passing schema and vocabulary checks therefore does not imply a non-overlapping segmentation. Activity vocabulary use ranges from 89.3% to 96.7%.

Table 11. Corpus and original transcript diagnostics. Activity and process columns show permitted/used labels. Coverage uses unions divided by parsed endpoint spans, not verified video duration. Alignment is boundary agreement within five seconds under the original parser.

Factory	Workers	Videos	Events	Activities	Processes	Coverage	Alignment
001	11	97	2,288	131/125	13/13	94.8%	93.1%
002	10	54	1,541	75/67	24/24	28.2%	71.8%
003	11	72	2,449	55/52	12/12	93.0%	86.8%
004	10	51	1,893	71/65	22/22	92.1%	90.4%
005	11	22	1,103	102/97	18/17	25.0%	75.0%
006	6	26	1,424	61/59	24/24	14.1%	62.1%
All	59	322	10,698			52.8%	82.1%

7.2. RQ2: Consistency with generated transcripts

The original parser finds 82.1% of event boundaries within five seconds of a recognized transcript endpoint. It yields an overall coverage diagnostic of 52.8%. Neither statistic is video accuracy. The coverage denominator is heavily affected by the literal multi-hour endpoints listed in Table 8. Alignment can also be low when formats change or when S4 splits a narrated interval. We remove the previous interpretation of low coverage as demonstrated extraction truncation of long videos.

Partial-support examples show the difficulty of checking events against compressed text. Event sample 5 assigns a particular bagging activity where the transcript only states that repetitive assembly and packaging continue. Sample 40 assigns guiding an upper mold half where the text only says that the worker helps with a mold. Recall sample 10 uses a transcript interval from 11:51:00 to 12:59:00 in an anomalous file whose event log ends at 00:20:00. Its MISS judgment is preserved, but the original explanation as long-recording truncation is unverified. These cases show why a single accuracy label would obscure different sources of uncertainty.

Table 12. Existing author judgments against generated transcripts. No additional audit was conducted.

Sample outcome	Count	Interpretation
Event fully supported (C)	36	Strict transcript support: 36/42, or 85.7%
Event partially supported (P)	6	Context is plausible; overlapping description is insufficient
Event contradicted (X)	0	No contradiction among the sampled events
Transcript entry overlaps (COV)	34	At least one event overlaps; action completeness not established
Uncovered, non-operational (EXC)	4	Excluded from the archive's reported overlap denominator
Uncovered, operational (MISS)	4	Reported entry-overlap rate: 34/38, or 89.5%

7.3. An end-to-end foundry example

Consider factory 006, worker 003, recording `factory006_worker003_00003`. The S1 transcript interval from 00:04:05 to 00:04:36 describes placing an empty frame on a pattern plate, pouring and hand-packing powder, and using a wooden tool to level the surface. The S2 summary describes preparation, filling, packing, leveling, and compression of molds. It compresses many recordings into a worker-level account. The S3 report supplies process *Sand mold creation*, activity *level powder surface with wooden scraper*, and a compact summary of this worker's molding work.

S4 produces two events within that 31-second transcript interval: *pack casting sand into mold frame*, from 00:04:05 to 00:04:24, and *level powder surface with wooden scraper*, from 00:04:24 to 00:04:36. Both have process *Sand mold creation*. Figure 5 shows what is retained and added. The internal boundary at 00:04:24 is inferred; S1 does not timestamp that transition separately. The events compress placement, pouring, and joint hand-packing into the first activity and add controlled names and a process context.

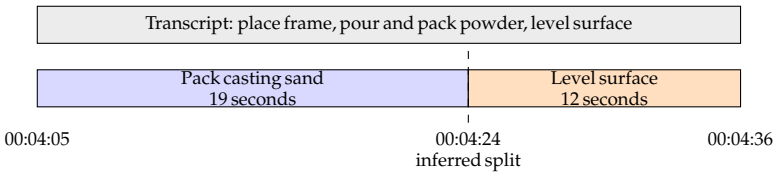


Figure 5. Exact transcript and event intervals for the foundry example. The original transcript is line 16 of its text file; the events are data rows 17 and 18 of the corresponding CSV. Bar lengths show recorded durations, not independently checked action boundaries.

```
{"events": [{
  "estimated_start_time": "00:04:24",
  "estimated_end_time": "00:04:36",
  "activity": "level powder surface with wooden scraper",
  "process": "Sand mold creation"
}]}
```

Listing 2: Illustrative JSON reconstructed from CSV data row 18. It is not an archived model response.

The source paths are `raw_transcriptions/factory_006/worker_003/factory006_worker003_00003.txt` and the corresponding path under `process_mining_event_logs/` with extension `.csv`. The summary is `worker_process_summaries/factory_006/worker_003.txt`; the vocabulary and compact context are in `factory_process_mining_reports/factory_006.txt`. S5 retains the source filename and local offsets alongside shifted timestamps. Audit event 41 judges the leveling event fully supported by its transcript. This

supports the label's textual basis, not the exact internal split or a video-based duration measurement.

7.4. RQ3: Descriptive analytical capabilities

7.4.1. Time composition and category transitions

Table 13 reproduces the existing category shares using the actual two-log exclusion rule. The non-VA category share ranges from 3.7% to 25.8% of retained summed event duration. This is neither total worker time nor recoverable capacity. Factory 003 assigns 10.8% to defect patching. Its factory report already describes defect patching in the opening overview and vocabulary. The contribution is to quantify that recorded behavior under an explicit category rule, not to discover previously unmentioned rework.

Table 13. Reproduced time analysis on 320 logs. Left: percentage of summed event duration. Right: factory medians over eligible workers, after the original worker-level rounding. A run sums consecutive same-category rows within a clip; transition counting can cross clips.

	Process-based category shares (%)							Worker summaries		
Factory	VA	MH	TR	QI	CM	RW	DOC	VA (%)	Trans./h	Mean run (min)
001	93.1	4.1	1.8	0.0	0.2	0.0	0.9	94.5	9.2	4.6
002	96.3	0.1	2.3	0.2	0.0	0.0	1.1	97.3	3.9	8.8
003	82.1	0.4	0.0	0.0	6.3	10.8	0.4	86.4	3.6	9.0
004	89.9	3.0	3.5	2.0	0.9	0.0	0.7	91.3	10.6	3.6
005	80.6	5.4	9.7	4.3	0.0	0.0	0.0	85.9	5.6	6.2
006	74.2	12.1	7.8	0.0	5.7	0.0	0.3	87.0	14.4	3.1

The mean-run column is the median across workers of their *mean* VA-run durations. A value of 3.1 minutes in factory 006 is a central tendency; it says nothing about the longest VA episode. Factory 001 has 449 MH/TR rows, a mean row duration of 14.54 seconds, and 14.6 such rows per summed event-hour. Factory 006 has 192 MH/TR rows, a mean of 25.14 seconds, and 28.4 rows per summed event-hour. These are row-based descriptions. Segmentation, repeated-action compression, and clip joins influence their values.

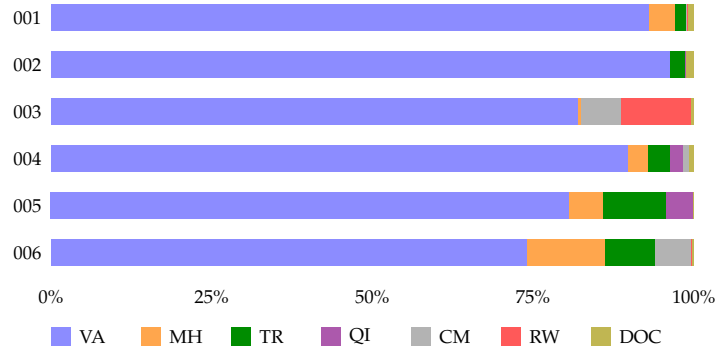


Figure 6. Category composition calculated from retained event rows and the released process map. Segments use unrounded duration totals; Table 13 reports rounded percentages. Factories have different task mixes and label schemes, so the plot does not rank efficiency.

7.4.2. Process flow and discovery

Video-as-case traces describe observation windows. Among the 97, 54, 72, 51, 22, and 26 videos in factories 001 to 006, there are respectively 97, 53, 70, 51, 22, and 26 distinct activity traces. Self-loops account for 4.3% to 16.8% of within-video directly-follows pairs. These statistics are reproduced directly from the stored labels. A self-loop can reflect separately extracted repetitions or segmentation choices; it is not a count of manufactured items.

The released discovery script uses Inductive Miner [20] with noise threshold 0.2, video cases, and activity labels. Appendix C records the repeated calculation and current dependency version. The recalculated fitness ranges from 0.9544 to 0.9910 and precision from 0.0603 to 0.2679. Fitness measures how the discovered model can replay the inferred traces; precision measures how much additional behavior it permits relative to that log. Both are properties of a model-log pair, not accuracy against actual factory behavior.

Many distinct traces and low precision can arise from flexible work, recording boundaries, abstraction choices, label variation, or extraction mistakes. The present evidence does not separate these causes. No process-level discovery result is reported because its computation and saved output are unavailable. Replacing activity labels with process labels would change the represented behavior and evaluation target. A higher precision after that change would not by itself be a controlled demonstration of better extraction or modeling. No new process-level models are generated for this revision.

7.4.3. Worker profiles and shared dominant processes

A shared vocabulary permits descriptive comparison within a factory. Table 14 groups workers by their largest process-duration total, then compares VA shares over *all* retained events for those workers. Eligibility requires at least 1,800 seconds of summed event duration. It does not require verified footage duration or equivalent tasks. There are 11, 10, 10, 10, 5, and 5 eligible workers in factories 001 to 006.

Table 14. Selected shared-dominant-process groups, using the original eligibility and rounding rules. Ranges summarize worker profiles; they are not estimates of individual efficiency.

Factory	Dominant process	Workers	VA share	Mean VA run
001	Manual mechanical assembly	6	89.7 to 97.8%	1.7 to 6.7 min
001	Metal stamping	3	90.6 to 98.5%	3.2 to 7.9 min
003	Surface finishing and polishing	5	67.3 to 96.9%	3.3 to 13.7 min
004	Garment ironing	2	52.8 to 77.1%	3.5 to 4.1 min
004	Overlock seaming	4	87.8 to 96.5%	1.5 to 10.3 min
005	Manual lathe machining	2	95.0 to 95.8%	6.9 to 9.4 min

The 29.6 percentage-point range in the surface-finishing group describes different observed profiles. Differences may reflect task mix, product requirements, recording selection, category assignment, or working conditions. Likewise, the small range for two lathe workers cannot rule out differences in behavior. Neither result supports individual performance rankings.

7.5. Six descriptive factory cases

Each case below states its recorded scope, quantitative result, relationship to the report, and unresolved interpretation. Factory descriptions are derived from the reports. Cross-factory rankings are limited by different vocabularies, products, and observation windows.

Factory 001: hardware stamping and assembly.

The archive has 97 clips from 11 workers; the time analysis retains 95 clips. Manual mechanical assembly accounts for 55.7% of retained summed duration, metal stamping for 18.2%, and component bagging for 11.3%. These differ from the earlier unfiltered dominant-process table because the two excluded logs are in this factory. The report already describes material replenishment, floor storage, and operator transport. The 449 MH/TR rows and their short mean duration quantify the assigned category pattern. They motivate examining material presentation, but do not establish that dedicated handlers, shorter travel, or a

different layout would improve output. Product mix and compressed repetitions remain competing explanations.

Factory 002: garment finishing and packaging.

There are 54 clips from ten workers, all retained by the event-end rule. Fabric folding, garment bagging, and ironing contribute 18.8%, 17.1%, and 13.2% of summed duration. VA is 96.3%. The report already describes transport, floor staging, and varied folding methods. Quantification shows a small separately categorized handling and transport share, not proof that those behaviors are rare within other processes. Two anomalous transcript endpoints and mixed formats weaken temporal interpretation. Product-specific finishing requirements and category boundaries need examination before any claim of superior efficiency.

Factory 003: surface preparation and assembly.

The archive covers 72 clips from 11 workers; all clips are retained. Surface finishing and polishing contributes 42.1%, wax pattern assembly 19.2%, and defect patching 10.8%. The report already identifies patching and material transport. The log quantifies patching's share without establishing defect rates, avoidable rework, or financial loss. It also records 101 transport events totaling 1,230 seconds even though the process-based TR share is zero. This case demonstrates how aggregation can hide an extracted behavior. Material types and quality criteria remain uncertain in the report itself.

Factory 004: garment sewing and finishing.

There are 51 clips from ten workers, all retained. Overlock seaming contributes 23.5%, garment folding 11.0%, and thread trimming 10.8%. The median worker VA-transition rate is 10.6 per summed event-hour. The report already describes tool switching, material handling, and inspection. The logs add a reproducible category pattern and a 24.3 percentage-point VA-share range in the two-worker ironing group. Product specifications, additional finishing duties, and recording windows could explain this difference. The data do not identify which changes to tools or stations would help.

Factory 005: machining and motor assembly.

The archive contains 22 clips from 11 workers, with only five workers meeting the time threshold. All clips are retained. Manual lathe machining contributes 32.1%, drill press machining 19.1%, and material transport and logistics 9.7%. The report describes staging, fixturing, inspection, and coupled workstations. The log adds duration shares, including 4.3% assigned to dimensional inspection. The two lathe profiles have similar VA shares, but their task equivalence is unknown. Inspection can be necessary, and one transcript has an unresolved endpoint anomaly. The recorded shares cannot measure excess quality work or recoverable machine capacity.

Factory 006: molding and foundry work.

There are 26 clips from six workers, all retained; five workers meet the time threshold. Sand mold creation, pattern cleaning and coating, and compression molding contribute 17.4%, 14.0%, and 12.2%. The report already describes floor staging, material movement, cleaning, and mold-track operations. The logs assign 19.8% of summed duration to MH/TR and yield a median worker mean VA run of 3.1 minutes. These quantities support a closer study of handling requirements, not a diagnosis of layout as their cause. Necessary mold preparation, differing materials, and segmentation choices are alternatives. Two transcript endpoints require resolution against source metadata.

7.6. From observations to operational hypotheses

Table 15. Evidence-linked hypotheses. These are candidates for investigation; intervention effects were not measured.

Observation	Plausible explanation	Competing explanation	Evidence needed
001: 449 short MH/TR rows; 5.9% of summed duration	Material presentation creates frequent replenishment	Segmentation splits one handling episode into several rows; task mix differs	Video checks, supply distances, batch quantities, station observations
006: 19.8% MH/TR; frequent category transitions	Layout or replenishment arrangements require repeated movement	Heavy or fragile materials require necessary support work	Routes, loads, safety requirements, matched task observations
003: 10.8% defect patching	Upstream defects require corrective work	Patching is a planned finishing requirement	Product specifications, defect criteria, item counts, upstream records
003 and 004: VA shares differ within dominant-process groups	Conditions or work allocation differ	Product mix, selection, or extraction differences	Comparable products and observation windows; independent video review
005: two similar lathe VA shares	Common station requirements shape profiles	Similar aggregate shares hide different tasks or behavior	Task-level comparisons and production context

Reports identify phenomena; logs can quantify them under a chosen representation. Explaining their causes requires additional evidence. Transport, inspection, cleaning, and documentation are not automatically waste, and the time assigned to them is not automatically available for production. No intervention effect or cost saving is demonstrated by these descriptive results.

8. Measurement and Analytical Implications

8.1. Observability and the interpretation of zeros

A zero in an analytical table has several possible causes (Table 16). These causes occur at different stages and require different checks. Requiring a label for transport or waiting improves representability, but does not establish reliable detection or a measured absence. A model may still omit the action, assign another label, or absorb it into a longer episode.

Table 16. Possible meanings of an apparent zero, ordered along the evidence path. Only the last mechanism is directly demonstrated by the factory 003 example; the others are possible mechanisms implied by the pipeline.

Mechanism	Where information is lost	Needed check
Behavior absent from the recorded window	Observation selection	Define the window and verify that it is suitable for the question
Behavior present but not narrated	Video interpretation or narration	Independent review of source footage
Behavior narrated but lost in summaries	S2 or S3 compression	Compare transcripts with worker and factory summaries
No suitable vocabulary label	Vocabulary induction	Inspect lists against the behavior of interest
Suitable label exists but is not assigned	Event extraction	Compare relevant transcript spans and extracted labels
Event exists but category aggregation hides it	Analytical mapping	Compare activity-level rows with process-derived categories

In factory 003, both *transport loaded tray to storage rack* and *transport bulk material via wheelbarrow* are permitted activity labels and occur in the logs. Together they account for 101 events and 1,230 seconds. No process label is mapped to TR in that factory. Their durations therefore inherit categories such as VA from their assigned processes. The TR

category's zero describes the map's output, not the absence of transport or the inability of the activity vocabulary to express it.

To interpret an observed zero as evidence of absence within a window, an analyst would need a defined observation scope, a suitable representation, adequate detection, correct assignment, and an aggregation rule that preserves the behavior. None follows from label membership alone. Conversely, an observed event can establish what the representation contains while leaving the event's visual accuracy unmeasured.

8.2. Uncertainty propagation and limits of provenance

Errors can propagate along both branches of Figure 3. A mistaken object description can shape the worker summary, the factory vocabulary, and the final event label. Agreement among those outputs can therefore reflect a shared error. Table 17 links each error source to an existing check and the conclusion it can affect.

Table 17. Error sources, existing checks, and analytical consequences. No calibrated stage-wise error rates are available.

Error source	Existing check and remaining gap	Possible analytical consequence
Visual interpretation	S1 asks for uncertainty; no independent video audit	Incorrect tools, materials, or actions enter all later outputs
Narration and repetition	Text can be inspected; no exhaustive action coverage check	Waiting or cycles disappear; episode counts change
Worker/factory summarization	Nonempty outputs and report parsing; no information-retention measure	Rare behavior does not influence the vocabulary
Vocabulary induction	Permitted lists are inspectable; completeness and compatibility are unchecked	No suitable label, or a forced approximate label
Event labeling	Membership checks and 42 archived judgments against text	A valid label can still describe the wrong action
Boundary estimation	Offset parsing, end-before-start checks, and transcript alignment; no source-duration check	Biased durations, overlap, or misleading endpoint spans
Case construction	Source files and offsets retained; real inter-clip order and gaps unknown	Artificial directly-follows relations and synthetic elapsed time
Analytical categorization	Public process map; no independent value classification	Supporting actions can be hidden inside VA contexts
Observation selection	Artifact inventory; selection protocol unavailable	Worker or factory differences may reflect sampling

Readable text is useful evidence, but it is weaker than structured event provenance. Existing records identify a source file and local interval. They do not store supporting transcript-span identifiers, alternative labels, calibrated confidence, or a flag that distinguishes an inferred internal boundary from a copied transcript endpoint. Finding the source of an event therefore requires reading and matching intervals. The foundry example can be traced manually, yet its CSV does not record why the split at 00:04:24 was chosen.

A future schema could add transcript-entry references, a boundary-origin field, vocabulary and prompt versions, alternative labels, and explicit uncertainty types [15]. Such fields would need evidence and validation. We do not retrofit invented confidence values to the existing events. Provenance can explain how a conclusion was reached; it does not certify that every preceding observation was correct.

8.3. Missing time does not imply a one-sided bound

The observed share outside VA is sometimes described as a lower bound because waiting and other supporting work can be omitted. That interpretation also assumes that omitted time is sufficiently non-VA. Missing VA time can instead raise the observed non-VA share.

Proposition.

Consider an idealized observation with $C > 0$ seconds of correctly classified, non-overlapping observed time, of which N seconds are non-VA. Let $U \geq 0$ seconds be unobserved. If observed and unobserved time partition a known target window, then its full-time non-VA share satisfies

$$\frac{N}{C+U} \leq p_{\text{nonVA}} \leq \frac{N+U}{C+U}. \tag{4}$$

Proof. Let x be the non-VA duration in the unobserved part. Since $0 \leq x \leq U$, the full share is $(N+x)/(C+U)$ and has the stated extrema. Both are attainable: all missing time may be VA or all may be non-VA.

For $U > 0$, the observed ratio N/C is a lower bound only if $x/U \geq N/C$. Thus, merely knowing that some waiting is missing is insufficient. The interval width in Equation 4 is $U/(C+U)$, which shows how unknown coverage limits a share even with perfect observed labels.

This is a conceptual result, not an empirical confidence interval for this dataset. Source durations and therefore U are unavailable; labels are inferred; and summed event durations include overlaps. Substituting the coverage diagnostic or the 97.1 retained summed event-hours into this proposition would violate its assumptions. The result identifies the additional evidence needed before making a bound claim.

8.4. Why aggregation changes the question

A duration-weighted factory share estimates the composition of the retained event rows. A median worker share gives equal weight to each eligible worker. They answer different questions. In factory 006, for example, the factory VA share is 74.2%, while the median worker VA share is 87.0%. Different observation amounts and work profiles can produce this difference without either statistic being erroneous.

Likewise, process-based and activity-based categories need not agree. Counting rows emphasizes the extraction granularity; counting runs suppresses adjacent labels; measuring interval unions removes overlap but does not resolve classification conflicts. None is a universal correction. A defensible analysis states its unit, denominator, case rule, handling of gaps, and interpretation before drawing operational conclusions. The released measurements serve as a reproducible example of those choices and their limits.

9. Discussion and Conclusions

9.1. Answers to the research questions

For RQ1, the five-stage pipeline produced a layered resource of transcripts, summaries, vocabularies, and 322 event logs. All 10,698 event rows satisfy the reported field, ordering, and label checks. Overlaps and unchecked semantic properties show why these checks must be named rather than treated as general validity.

For RQ2, the existing audit finds full transcript support for 36 of 42 sampled events and partial support for six. Boundary agreement and entry overlap add temporal evidence, but format inconsistencies limit their interpretation. Accuracy against source footage remains unmeasured. The audit does not bound total downstream uncertainty.

For RQ3, the retained logs support descriptions of category composition, local sequences, and worker profiles. They quantify an already described defect-patching process and expose differences worth investigating. The evidence supports operational hypotheses, not causal diagnoses, individual efficiency scores, or expected savings. Qualitative reports and quantitative logs complement one another but share generated transcripts.

For RQ4, the examples establish three general lessons. Recorded behavior can disappear under an aggregation rule. Synthetic case and time construction can introduce transitions that were not observed continuously. Missing time can move a category share in either direction. These lessons apply to inferred logs beyond video whenever an analysis depends on abstraction, selection, and reconstructed timestamps.

9.2. Limits tied to the conclusions

The release lacks the historical source revision, selection protocol, duration metadata, full model execution records, and complete sampling code. These gaps limit exact regeneration and claims about representative factory behavior. The frozen artifacts still permit the descriptive calculations to be reproduced, and the revision notes identify the missing historical information explicitly.

The process map is a coarse analytical convention. It can assign transport to VA and combines necessary support work with activities an analyst might consider avoidable. It cannot estimate waste or recoverable production capacity. Product identifiers, output counts, quality specifications, and material links across workers are absent, so the logs cannot establish throughput, first-pass yield, upstream defect rates, or complete product routes.

Temporal inconsistencies affect endpoint coverage; overlaps affect summed durations; gap-bridging runs affect fragmentation language; and clip joins affect transitions. The event-end exclusion rule leaves some inconsistent transcripts in the analysis. The retained numerical tables should therefore be interpreted as descriptions of frozen representations, with their sensitivity to unresolved time errors still unknown. We do not convert anomalous timestamps to more plausible values without source evidence.

The author audit is small, has one evaluator, and refers to generated text. It cannot detect transcription errors shared by the report and log. Unequal windows and dominant-process grouping do not establish matched worker tasks. Alternative explanations in Table 15 remain open, and no intervention was evaluated. These limitations directly restrict operational conclusions rather than merely limiting generalization to other factories.

9.3. Responsible interpretation and use

Worker-level process data can be turned into individual profiles even when names are removed [21]. The release uses source worker identifiers and does not redistribute video. These identifiers are pseudonymous, not a guarantee against re-identification. A distinctive task, timestamp pattern, or combination with local records may reveal a person or role.

The intended use is research on extraction and measurement, and investigation of process design at an appropriate aggregate level. The profiles should not be used for employee appraisal, pay, discipline, or automated worker ranking. The unequal observation windows and inferred category assignments do not support those decisions. For workplace use, access should be restricted to the stated analytical purpose, with aggregation across suitable task groups and review with the people whose work is represented. Results should retain their coverage and provenance limits when shared.

The provider describes real-factory collection and publishes a dataset card and access terms [3]. The inspected card does not document the original consent procedure. The present authors performed secondary processing, collected no new video, and report no new consent or ethics-approval record. Provider statements, author processing choices, and unavailable collection details are therefore kept separate. The pipeline prompts discourage worker-level judgments, but prompts alone do not control later use of the public outputs.

9.4. Reusable contribution and future work

The resource enables two different activities: reproducing calculations from fixed outputs, and regenerating outputs through model APIs. The first is portable and documented in Appendix B. The second depends on source access, model availability, and historical settings that are only partly recorded. Exact regeneration is therefore not claimed.

Future work could preserve video-duration metadata, add explicit activity-process compatibility, link events to transcript spans, and audit against video with multiple evaluators. Windowed extraction and alternative category rules are design candidates, not validated improvements. Real calendar times and object references would be needed for synchronized flow and product-level analysis.

The paper's contribution is a concrete method, an inspectable data resource, and a careful account of what its measurements mean. The logs make selected manual work available for structured study. Their usefulness depends on retaining the evidence path and matching each conclusion to the properties that have actually been assessed.

Declarations

Author Contributions: Alessandro Berti carried out the experiments. Wil M. P. van der Aalst reviewed the manuscript and suggested improvements.

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Institutional Review Board Statement: This study is a secondary analysis of the existing Egocentric-10K dataset. It involved no recruitment of participants, intervention, or new video collection. No ethics approval or exemption record for this secondary analysis is reported.

Informed Consent Statement: No participants were recruited or contacted for this secondary analysis. The source videos were obtained from the existing Egocentric-10K dataset. The authors did not collect new consent, and the source dataset card does not report the original consent procedure.

Data Availability Statement: The pipeline, prompts, transcripts, factory vocabularies, derived event logs, and assessment code are available at https://github.com/fit-alessandro-berti/annotated-egocentric-10k-dataset. The source videos are available from Build AI's Egocentric-10K dataset at https://huggingface.co/datasets/builddotai/Egocentric-10K, under its access terms and Apache License 2.0. The repository contains derived artifacts and does not redistribute the source videos.

Acknowledgments: The study used Gemini 3.1 Pro for video descriptions, worker summaries, and factory reports, and GPT-5.4 for event extraction, as described in the methods. OpenAI Codex (GPT-6) assisted with the journal revision, evidence checks, descriptive recalculation scripts, language editing, LaTeX formatting, and preparation of the cover letter. The authors are responsible for the content of this work.

Conflicts of Interest: Wil M. P. van der Aalst is affiliated with Celonis, a process mining software company. No other conflicts of interest are reported.

Appendix A. Journal Extension and Evidence Record

Table A1 compares the supplied short manuscript with this revision. It documents the extension of that manuscript; it does not assert a publication status for a workshop version. No new extraction experiment or audit has been added. The complete task-to-deliverable and claim-to-evidence registers are in `paper/revision/`.

Table A1. Workshop-to-journal revision map relative to the supplied short manuscript.

Existing material	Journal addition	Type of contribution
Five-stage overview	Formal event model, implementation algorithm, stage failures, guarantee table	Method specification
Brief time description	Synthetic-time equation, case comparison, timestamp and inclusion manifests	Measurement analysis and reporting correction
Category percentages	Activity/context/category distinction and exact transport example	Methodological finding from existing artifacts
Coverage caveats	Missing-time proposition with explicit assumptions	New analytical reasoning
Short foundry illustration	Transcript timeline, reconstructed event, provenance chain	Evidence-traceability example
Condensed audit	Rubric, denominators, ambiguity examples, all judgments	Fuller reporting of existing assessment
Numerical results	Portable recalculation, manifests, corrected terminology, six cases	Reproducibility and interpretation
Brief limitations	Observability taxonomy, uncertainty path, operational hypotheses, responsible use	Methodological synthesis

Appendix B. Reproducibility and Configuration

The offline command below reads the five frozen artifact directories and the original category map. It writes derived CSVs, a JSON summary, and typeset appendices to a separate output directory. It requires Python 3 and no model credentials or third-party packages.

```
python3 paper/reproduce.py
# Optional explicit locations:
python3 paper/reproduce.py --root /path/to/repository \
  --output-dir /path/to/results
```

The output includes `artifact_manifest.csv` (path, size, SHA-256), `inclusion_manifest.csv` (all 322 decisions), `transcript_intervals.csv`, `structural_metrics.csv`, `factory_metrics.csv`, `worker_metrics.csv`, `worker_comparisons.csv`, `factory003_transport.csv`, and `audit_judgments.csv`. The summary records script and prompt hashes. Merged-log consistency is checked separately by reconstructing S5 output in memory and comparing it to every released merged file. Source artifacts remain unchanged.

The original pandas-based assessment and value-analysis scripts accept portable output paths. The original activity-level discovery script retains its miner, noise threshold 0.2, and video case rule. Running it requires the dependencies documented in `paper/requirements-assessment.txt`. The environment used for this revision is recorded in `paper/revision/assessment_environment.txt`; it is not claimed to be the original environment. Table A2 maps paper results to their calculations.

Regeneration is a separate workflow using scripts 01 to 05 in order. Paths and model identifiers are command-line options in the extraction scripts. S1 requires source video archives and a Gemini credential; S2 and S3 use saved text and Gemini; S4 uses saved text, vocabularies, and an OpenAI credential; S5 is local. Their default model strings are `gemini-3.1-pro-preview` and `gpt-5.4`. S4 configures high reasoning effort and a strict JSON response schema; S1 to S3 do not explicitly configure temperature, a seed, or generation-token limits. Transient HTTP status codes 408, 429, 500, 502, 503, and 504 trigger retries; the scripts configure up to five attempts with increasing delay. S1 also has an outer per-video retry loop. These are code settings, not verified execution totals.

The archive does not provide per-request model responses, usage records, exact execution dates, token counts, costs, runtimes, historical dependency versions, or a re-

Table A2. Reproduction map. Paths for the original scripts are relative to `paper/docs/assessment_materials/`.

Paper result	Calculation and frozen input	Saved evidence
Tables 7, 8, 11	`reproduce.py`; source texts and per-video logs; `assess.py` for original diagnostics	Artifact, inclusion, interval, and structural manifests
Tables 13, 14 and factory cases	`reproduce.py` and `value_analysis.py`; original process map and retained rows	Factory metrics, worker metrics, comparisons, composition totals
Figure 6	`reproduce.py`; exact duration totals and original map	Generated TikZ bars and category percentages
Figure 2 and Section 8.1	Activity-label filter over factory 003 CSVs	101 rows in `factory003_transport.csv`
Table 12, Appendix E	Mechanical transcription of `audit_verdicts.md`	84 judgments in CSV and LaTeX
Section 7.4.2	Original `pm_utility.py`, all per-video rows	`activity_discovery.json` and execution output
Equation 4	Algebra under explicitly idealized assumptions	Proof in Section 8.3; no empirical bound

solved source revision. Existing outputs may be reused without a content-validity check. Consequently, changing a prompt while retaining existing outputs does not prove that those outputs were generated with the changed prompt. Frozen artifacts, hashes, and the remaining historical queries are recorded so that this limitation is explicit.

Appendix C. Activity-Level Discovery Recalculation

Table A3. Recalculation with the original activity-level discovery settings. Fitness is average trace fitness from token replay; precision is token-based precision.

Factory	Cases	Fitness	Precision
001	97	0.9801	0.0691
002	54	0.9910	0.0603
003	72	0.9874	0.0906
004	51	0.9823	0.0666
005	22	0.9544	0.2679
006	26	0.9720	0.0832

PM4Py version for this recalculation: 2.7.23.11. The original settings are retained. The saved result records the current PM4Py version. No archived process-level computation or result was found, so no process-level fitness or precision value is reported. Discovery-model images are omitted because the original models were not preserved. The repeated calculation establishes the shown model-log metrics under the recorded current environment, not a recovery of the original runtime.

Appendix D. Complete Operational-Category Map

This table reproduces the original map with factory scope made explicit. VA denotes production/completion work under the authors' convention, MH material handling, TR transport, QI inspection, CM cleaning/maintenance, RW rework, and DOC documentation. The sole unmapped label is unused in the archived events.

Factory	Process label	Category
001	Metal stamping	VA
001	Metal forming	VA
001	Manual mechanical assembly	VA

Factory	Process label	Category
001	Semi-automated press assembly	VA
001	Structural frame assembly	VA
001	Component bagging	VA
001	Component kitting	VA
001	Material replenishment	MH
001	Finished goods transport	TR
001	Scrap disposal	MH
001	Workstation cleaning	CM
001	Production tracking	DOC
001	Machine maintenance	CM
002	Fabric untangling	MH
002	Fabric edge alignment	VA
002	Pattern matching	VA
002	Fabric folding	VA
002	Fabric pairing	VA
002	Fabric pinning	VA
002	Sub-assembly marking	VA
002	Fabric bundling	VA
002	Garment ironing	VA
002	Garment quality inspection	QI
002	Thread snipping	VA
002	Garment tagging	VA
002	Hanger assembly	VA
002	Garment bagging	VA
002	Cardboard sleeve packaging	VA
002	Cardboard wrapping	VA
002	Package labeling	VA
002	Master bagging	VA
002	Box assembly	VA
002	Box packing	VA
002	Box sealing	VA
002	Cart loading	TR
002	Material transport	TR
002	Floor auditing	DOC
003	Metal casting separation	VA
003	Wax pattern assembly	VA
003	Surface finishing and polishing	VA
003	Defect patching	RW
003	Component sub-assembly	VA
003	Injection molding	VA
003	Mold preparation and cleaning	CM
003	Machine pressing	VA
003	Spot painting	VA
003	Thermal welding	VA
003	Material sorting and sifting	MH
003	Inventory verification	DOC
004	Fabric spreading	VA
004	Bulk fabric cutting	VA
004	Fabric measuring	VA
004	Overlock seaming	VA
004	Continuous fabric chaining	VA
004	Edge binding attachment	VA

Factory	Process label	Category
004	Piping attachment	VA
004	Curvilinear seam joining	VA
004	Tape feeding	VA
004	Heat press labeling	VA
004	Garment hemming	VA
004	Thread trimming	VA
004	Quality inspection	QI
004	Garment folding	VA
004	Garment ironing	VA
004	Garment bagging	VA
004	Bulk box assembly	VA
004	Pick-and-pack sorting	MH
004	Production data logging	DOC
004	Scrap material management	MH
004	Machine maintenance	CM
004	Material transport	TR
005	motor assembly finishing	unmapped
005	motor labeling and nameplate attachment	VA
005	motor packaging and strapping	VA
005	automated stator winding	VA
005	semi-automatic stator wire insertion	VA
005	stator coil binding	VA
005	stator lead wire finishing and soldering	VA
005	manual lathe machining	VA
005	CNC machine tending	VA
005	drill press machining	VA
005	milling machine operation	VA
005	component deburring and polishing	VA
005	dimensional quality inspection	QI
005	manual hardware pre-assembly	VA
005	batch pneumatic fastening	VA
005	material transport and logistics	TR
005	part sorting and staging	MH
005	finished goods packaging	VA
006	Sand mold creation	VA
006	Pattern cleaning and coating	VA
006	Core and insert placement	VA
006	Mold assembly and closing	VA
006	Mold track advancement	MH
006	Machine cleaning and maintenance	CM
006	Manual production logging	DOC
006	Wooden component scraping	VA
006	Wooden component assembly	VA
006	Machine sanding	VA
006	Powder sifting	VA
006	Chemical dough mixing	VA
006	Dough kneading	VA
006	Machine extrusion	VA
006	Punch press operation	VA
006	Bulk material transport	TR
006	Component staging and arrangement	MH
006	Floor-based part sorting	MH

Factory	Process label	Category
006	Part mechanical modification	VA
006	Compression molding	VA
006	Raw material mixing	VA
006	Post-processing edge finishing	VA
006	Metal casting separation	VA
006	Workspace clearing	CM

Appendix E. Complete Archived Audit Judgments

The following tables preserve all 84 verdicts and source identifiers. The shorthand `001/w003/v00009`, for example, identifies factory 001, worker 003, recording 00009. Event verdicts are C (fully supported), P (partially supported), and X (contradicted). Entry verdicts are COV (overlap), EXC (uncovered and judged non-operational), and MISS (uncovered and judged operational). The archive's unsupported truncation explanation is flagged in entry 10. Punctuation is normalized for typesetting; judgments are unchanged.

Appendix E.1. Archived event judgments

No.	Source	Interval; verdict	Label and archived note
1	001/w003/v00009	00:01:38-00:02:12; C	mate components over central jig; assembly over metal block
2	001/w002/v00001	00:08:41-00:08:48; C	drop sealed bags into bulk container
3	001/w006/v00000	00:07:50-00:08:02; C	receive part from adjacent coworker
4	001/w005/v00006	00:17:09-00:17:15; P	actuate machine press; interval covers full load-press-remove cycle
5	001/w005/v00004	00:14:00-00:14:55; P	place main component into larger bag; transcript: "repetitive assembly and packaging continues"
6	001/w004/v00000	00:06:34-00:06:43; P	drive fastener into metal parts; transcript: "repetitive assembly continues"
7	001/w003/v00006	00:04:15-00:06:22; C	insert small pin into base components; split of long composite entry
8	002/w009/v00000	00:19:07-00:20:00; C	align printed fabric patterns
9	002/w006/v00002	00:05:39-00:06:11; C	enclose garment in pre-printed cardboard folder
10	002/w002/v00007	00:10:06-00:10:18; C	attach black plastic hanger hook to bag
11	002/w007/v00002	00:08:16-00:08:31; C	wrap white cardboard band around folded shirt
12	002/w005/v00007	00:06:09-00:06:24; C	fold t-shirt over removable white template; split of repeated-cycle entry
13	002/w002/v00001	00:12:22-00:12:26; P	drop packaged garments into bulk box; transcript: "appearing to drop an item"
14	002/w002/v00001	00:11:41-00:11:43; C	drop packaged garments into bulk box
15	003/w003/v00000	00:10:36-00:13:22; C	scrape component surface with metal tool
16	003/w005/v00004	00:04:13-00:04:29; C	strike metal casting with hammer
17	003/w005/v00004	00:15:47-00:15:53; C	scoop loose parts from bulk bin
18	003/w010/v00000	00:11:45-00:12:05; C	squeeze binding paste from piping bag
19	003/w001/v00001	00:05:45-00:06:12; C	fill metal gap with dark clay
20	003/w010/v00003	00:17:12-00:17:29; C	load pre-manufactured insert into mold
21	003/w005/v00000	00:03:59-00:07:18; C	strike metal casting with hammer
22	004/w007/v00001	00:14:07-00:14:36; C	Iron folded garment with steam iron
23	004/w006/v00004	00:16:20-00:16:29; C	Unfold sewn piece to inspect seam tension
24	004/w007/v00001	00:07:22-00:07:24; C	Snap large cardboard sheet in half
25	004/w006/v00002	00:17:00-00:17:03; C	Snip loose threads with small clippers
26	004/w004/v00005	00:14:08-00:14:16; C	Push finished fabric chain to table edge
27	004/w003/v00000	00:00:00-00:01:00; C	Hem large fabric piece in short bursts; merges 5 s stand-up interruption
28	004/w005/v00001	00:08:44-00:15:18; C	Sew short straight seam; aggregated repetition
29	005/w004/v00002	00:04:43-00:04:51; C	brush metal shavings off lathe tool post
30	005/w001/v00000	00:06:26-00:06:36; C	position metallic nameplate on motor housing; split of ten-motor loop entry
31	005/w003/v00000	00:08:20-00:08:27; C	advance cutting tool via carriage handwheels
32	005/w008/v00001	00:19:10-00:19:18; P	arrange parts manually inside deep wire mesh bin; tail of composite transport entry
33	005/w006/v00000	00:06:03-00:06:10; C	initiate milling machine cycle via control panel
34	005/w003/v00000	00:07:22-00:07:29; C	advance cutting tool via carriage handwheels
35	005/w004/v00000	00:02:23-00:02:33; C	apply adhesive label to motor housing

No.	Source	Interval; verdict	Label and archived note
36	006/w003/v00002	00:12:28-00:12:38; C	lift empty mold box onto pattern plate; "onto the machine"
37	006/w001/v00004	00:05:08-00:06:01; C	sift through mixed small parts on floor; floor bin
38	006/w001/v00002	00:08:42-00:08:58; C	guide upper mold half onto lower mold half
39	006/w003/v00003	00:11:39-00:11:55; C	pack casting sand into mold frame
40	006/w001/v00002	00:11:00-00:11:07; P	guide upper mold half onto lower mold half; transcript vague: "helps with a mold"
41	006/w003/v00003	00:04:24-00:04:36; C	level powder surface with wooden scraper
42	006/w003/v00002	00:16:15-00:16:30; C	pack casting sand into mold frame

Appendix E.2. Archived transcript-entry judgments

No.	Source	Interval; verdict	Label and archived note
1	001/w008/v00010	00:12:00-00:12:51; COV	continues repetitive assembly
2	001/w004/v00007	00:05:22-00:05:39; COV	tears open bag of new metal parts
3	001/w002/v00003	00:01:07-00:01:12; COV	staples bag shut, sets aside
4	001/w010/v00002	00:11:22-00:12:05; COV	aligns hinged-clip rail on plate in jig
5	001/w007/v00004	00:15:43-00:16:41; COV	resumes stamping operation
6	001/w008/v00002	00:08:46-00:08:49; COV	camera tilt (overlapped by longer event)
7	001/w003/v00005	00:09:10-00:09:15; COV	places assembled parts onto cardboard
8	002/w005/v00009	00:20:10-00:20:13; COV	places cardboard piece on t-shirt
9	002/w003/v00000	00:09:16-00:09:44; COV	fetches and aligns garment stacks
10	002/w010/v00001	11:51:00-12:59:00; MISS	aligning stack of pink fabric pieces (archive interprets this as truncation; that cause is unverified)
11	002/w005/v00003	00:03:38-00:03:47; COV	irons folded fabric
12	002/w005/v00009	00:09:44-00:09:49; COV	irons t-shirt back
13	002/w004/v00006	00:13:17-00:13:20; EXC	short walking sequence
14	002/w002/v00009	00:11:01-00:12:35; COV	two workers fill sack with garments
15	003/w001/v00002	00:18:12-00:18:42; COV	continues scraping task
16	003/w009/v00003	00:19:35-00:19:42; COV	empties box onto tray
17	003/w004/v00002	00:02:58-00:03:08; COV	applies white paste with brush
18	003/w004/v00007	00:19:45-00:19:55; COV	continues filing metal part
19	003/w002/v00000	00:18:30-00:18:50; MISS	fetches empty tray, stages tied bundles
20	003/w004/v00002	00:06:27-00:06:33; COV	brushes paste onto part
21	003/w002/v00004	00:02:48-00:02:51; COV	drops debris into box
22	004/w006/v00004	00:17:28-00:17:34; COV	checks seam, folds fabric
23	004/w005/v00003	00:00:40-00:00:48; COV	prepares fabric edge for sewing
24	004/w009/v00001	00:09:44-00:09:46; COV	drops packaged stack in box
25	004/w006/v00004	00:08:12-00:08:29; COV	aligns fabric, resumes sewing
26	004/w003/v00000	00:19:50-00:20:00; COV	resumes continuous sewing
27	004/w006/v00004	00:11:38-00:11:43; EXC	checks phone briefly
28	004/w006/v00004	00:01:24-00:01:40; COV	sews another section
29	005/w002/v00000	00:08:04-00:08:33; EXC	brief walk around workbench, returns
30	005/w008/v00000	00:01:35-00:01:42; EXC	camera shake while walking
31	005/w003/v00000	00:05:32-00:05:40; COV	tightens fixture with wrench
32	005/w008/v00000	00:11:36-00:11:45; MISS	standing, inspecting part (borderline operational)
33	005/w011/v00000	00:08:56-00:09:22; MISS	presses control-panel buttons, consults sheet
34	005/w008/v00000	00:05:32-00:05:54; COV	lifts bin with pallet jack, transports
35	005/w006/v00001	00:02:00-00:04:00; COV	workers stack red components into wire basket
36	006/w001/v00002	00:08:58-00:09:12; COV	waits with spray gun, sprays mold
37	006/w007/v00000	00:03:13-00:03:22; COV	hammers casting apart
38	006/w007/v00000	00:15:10-00:15:23; COV	hammers casting, scrapes sand
39	006/w001/v00007	00:11:57-00:12:03; COV	blows air onto pattern plate
40	006/w005/v00001	00:02:47-00:02:55; COV	carries mold, operates control panel
41	006/w001/v00008	00:19:19-00:19:32; COV	flips pattern plate, air-cleans
42	006/w001/v00007	00:09:24-00:09:35; COV	two workers move sand mold with lifting clamp

Appendix F. Full Extraction Prompts

These are the four released prompt templates, with punctuation normalized for display. The source files and their hashes remain in the resource. They specify requested model behavior, not guaranteed output properties. Placeholder braces are filled by the scripts; additional context is constructed as described in Table 4.

Appendix F.1. S1: Video transcription

```
Please transcribe the attached egocentric video into a very detailed chronological text description.
```

```
I do not want a formal event log, structured schema, activity labels, or any kind of post-processing. I only want plain text entries, each beginning with a timestamp range, followed by a detailed description of what happens in that time span.

Requirements:
- Format every entry as:
  [HH:MM:SS - HH:MM:SS] description
- Follow the video strictly from start to end.
- Use short timestamp intervals whenever something changes.
- Be highly detailed, but do not repeat the same type of behavior again and again when nothing changes.
- If the same action continues in the same way, with the same context and no relevant difference, merge it into one longer timestamped entry instead of splitting it into many repetitive entries.
- Only create a new entry when something meaningful changes, such as:
  - the action changes
  - a new object is handled
  - the person changes location
  - attention shifts to something else
  - another person becomes involved
  - something unusual, problematic, interrupted, or different happens
- Capture micro-actions when visible, such as reaching, grasping, lifting, placing, opening, closing, turning, walking, pausing, looking, adjusting, and interacting with objects or people.
- Describe what is visible in the egocentric view, including hand movements, object handling, scene changes, and shifts of attention.
- If a repeated behavior continues in exactly the same way, summarize that repeated stretch once with an appropriate timestamp range instead of listing each repetition separately.
- If the scene is partially unclear, mention the uncertainty briefly instead of guessing.
- Do not use tables, bullets, JSON, labels, categories, or analysis language.
- Do not convert the result into an event log.
- Do not add any summary at the end.
- Only provide the timestamped descriptive transcription.

The output should read like a detailed time-aligned narration of the video, not like a structured annotation file.

Example of the intended compression:
- Good:
  [00:03:10 - 00:03:42] The wearer continues wiping the same section of the table with repeated back and forth hand movements. No new object or interruption appears.
- Not desired:
  [00:03:10 - 00:03:14] The wearer wipes the table.
  [00:03:14 - 00:03:18] The wearer continues wiping the table.
  [00:03:18 - 00:03:22] The wearer continues wiping the table.

Please produce the full transcription in this style for the entire video.
```

Appendix F.2. S2: Worker summary

```
You are given multiple detailed chronological transcriptions from egocentric factory videos recorded by the same worker.

Your task is to synthesize those transcriptions into a process-focused report about what work is being done and what process-level optimization opportunities are visible.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

Important output constraints:
- Do not mention the factory identifier, worker identifier, file names, file paths, transcript numbers, video numbers, or any other source identifiers.
- Do not mention timestamps unless they are absolutely necessary, which they normally are not.
- Do not quote the transcripts.
- Do not write a per-video recap. Synthesize across all provided transcriptions.

Required focus:
- Identify the process or processes performed by the worker.
- If the worker appears to perform more than one distinct process, separate them clearly.
- Describe the operational flow in enough detail that someone can understand how the work is carried out.
```

```
- Include relevant details such as setup, repeated action loops, machine interaction, material handling,
 part movement, quality checks, cleaning, waiting, transport, handoffs, and interaction with nearby
stations when visible.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly
instead of inventing details.

Optimization constraints:
- Only suggest process optimizations that are visible or reasonably inferable from the observed workflow.

- Focus on process design, workstation layout, batching, staging, transport, sequencing, buffering,
machine idle time, tool/material presentation, inspection placement, handoffs, or coordination between
steps.
- Do not suggest worker-level optimizations such as "work faster", "be more careful", "improve
discipline", "train the worker better", or similar people-focused advice.
- If no credible process optimization is evident, say so.
- When an optimization is inferential rather than directly observed, label it clearly as an inference.

Write the output in plain text with exactly these section headings:

Process Summary
Detailed Process Breakdown
Possible Process Optimizations
Evidence Limits
```

Appendix F.3. S3: Factory report and vocabularies

```
You are given multiple worker-level process summaries from the same factory.

Your task has two goals:
1. Produce a factory-level report that restates the worker process summaries in a compact but
informative way, preserving which worker each summary belongs to.
2. Produce a long, detailed, process-mining-oriented activity catalogue for the factory that can help
with downstream process optimization.

Context:
- Factory identifier: {factory}

Important constraints:
- Do not mention file names, file paths, source identifiers, transcript numbers, or video numbers.
- Do not quote the source summaries.
- Stay grounded in the provided evidence. If something is uncertain, state the uncertainty briefly
instead of inventing details.
- Synthesize across workers. Do not merely copy the worker summaries back verbatim.

Worker summary requirements:
- Include every worker provided in the input.
- Keep each worker summary concise but useful.
- Preserve the worker identifier for each summary, because the output is factory-level.
- Focus on the main process(es), major material flows, handoffs, machine interactions, transport,
inspection, waiting, cleaning, batching, and coordination patterns that matter operationally.

Factory process label requirements:
- Explicitly provide one raw deduplicated list of processes executed across the different workers in the
 factory.
- Express each process as a short label only.
- Keep process labels concise and comparable across the factory.
- Prefer labels such as "metal stamping", "box assembly", "garment bagging", "part transport", "
inspection", or "machine cleaning".
- Distinguish clearly between materially different processes, but do not create unnecessary micro-
variants.
- Do not group the labels by worker.
- Do not repeat the same label for multiple workers.

Activity catalogue requirements:
- The catalogue must be long and detailed.
- Each activity must be a short string only, suitable as a candidate event label for process mining.
- Prefer specific verb-object phrases such as "load raw blank into press" rather than generic labels
such as "operate machine".
- Include not only value-adding operations, but also transport, staging, waiting, inspection, rework-
like handling, setup, cleanup, handoff, documentation, and coordination activities when visible.
- Keep distinct activity variants separate when they matter for process analysis.
- Avoid duplicate or near-duplicate labels.
```

```
- Avoid full sentences, explanations, numbering inside the activity text, or extra commentary next to
the activity.
- If an activity is not clearly supported, do not invent it.

Optimization relevance requirements:
- Favor activity labels that would help reveal bottlenecks, queues, machine idle time, material handling
 burden, unnecessary motion, batching behavior, rework, inspection points, coordination dependencies,
and handoff structure.

Write the output in plain text with exactly these section headings:

Factory Process Overview
Worker Process Summaries
Factory Process Labels
Factory Activity Catalogue
Optimization-Relevant Process Observations
Evidence Limits

Formatting rules for the output:
- Under "Worker Process Summaries", use one line per worker in the form:
  worker_xxx: summary
- Under "Factory Process Labels", use one bullet per process label.
- Each bullet in "Factory Process Labels" must contain only a short process label and nothing else.
- Under "Factory Activity Catalogue", use one bullet per activity.
- Each bullet in "Factory Activity Catalogue" must contain only a short activity string and nothing else.
```

Appendix F.4. S4: Event extraction

```
You are converting one timestamped raw factory annotation into a structured event list for process
mining.

Context:
- Factory identifier: {factory}
- Worker identifier: {worker}

You will be given:
- the allowed process labels for the factory
- the allowed activity labels for the factory
- an optional worker summary for additional context
- one raw annotation transcript with timestamp ranges measured from the start of the video

Your job:
- Convert the raw annotation into a JSON object with one key, ‘events‘, whose value is a list of event
objects.
- Each event must correspond to one operationally meaningful event segment.
- Use the transcript timestamps as the main basis for the event boundaries.
- If a transcript interval contains one repeated stable activity, keep it as one longer event.
- If a transcript interval clearly contains multiple different events, you may split it into multiple
estimated events.
- Keep the events in chronological order.

Strict label constraints:
- The value of ‘activity‘ must be exactly one label from the allowed factory activity list.
- The value of ‘process‘ must be exactly one label from the allowed factory process list.
- Do not invent new activity labels.
- Do not invent new process labels.
- If multiple labels are plausible, choose the closest allowed label.
- Be especially careful to respect these two controlled vocabularies exactly.

Output schema:
- Return JSON only.
- Return a top-level JSON object with exactly one key: ‘events‘.
- The value of ‘events‘ must be a JSON array.
- Each array item must be an object with exactly these keys:
  - ‘estimated_start_time‘
  - ‘estimated_end_time‘
  - ‘activity‘
  - ‘process‘
- ‘estimated_start_time‘ and ‘estimated_end_time‘ must be strings in ‘HH:MM:SS‘ format, measured from
the start of the video.
```

```
Additional rules:
- Include only process-relevant operational events. Ignore purely accidental camera motion or camera
removal unless it directly reflects a process-relevant activity.
- Do not output markdown fences.
- Do not output explanations, notes, comments, or any text before or after the JSON object.
- If no valid operational events can be extracted, return `{"events":[]}`.
```

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